# How well do charmonium and bottomonium determine the $B_c$ spectrum?

A potential-model test with the $B_c(2S)$ hyperfine difference

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We ask to what extent the spectrum of the $B_c$ meson is fixed by charmonium and bottomonium data alone. Three families of spin-independent potentials—Cornell, logarithmic, and a power law with variable confining exponent $\alpha$—are refitted, together with the quark masses and the strong coupling, to 27 $J$-resolved $c\bar{c}$ and $b\bar{b}$ levels from the 2024 Review of Particle Physics, with spin-dependent splittings treated in first-order perturbation theory. Without any $B_c$ input the predictions for $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ agree across families to $\pm 15$ MeV, while $P$-, $D$- and higher $S$-wave levels show 20–150 MeV model dependence that we quantify as a function of $\alpha$. The $c\bar{c}+b\bar{b}$ data strongly favour a nearly logarithmic potential ($\alpha \lesssim 0.1$). The difference of hyperfine splittings $\Delta_{1S}-\Delta_{2S}$, which CMS and LHCb measure at 29–31 MeV, is predicted at 28–40 MeV and independently selects $\alpha \simeq 0.1$–$0.2$. With a fixed $\alpha_s$ the absolute $1S$ hyperfine splitting comes out at 76–82 MeV, well above the $64.5 \pm 1.7$ MeV recently measured by ATLAS; a one-loop running coupling with $\Lambda$ determined by $c\bar{c}+b\bar{b}$ alone predicts $\Delta_{1S} = 63$–$72$ MeV and $\Delta_{2S} = 34$–$38$ MeV, reproducing the ATLAS, CMS and LHCb hyperfine measurements simultaneously without any $B_c$ input. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV above the measured value in every variant. We also give $E1$ and $M1$ radiative widths for the lowest excited $B_c$ states.

## 1 Introduction

The $B_c$ meson is the only hadron built from two heavy quarks of different flavour. It is therefore both a laboratory for heavy-quark dynamics intermediate between charmonium and bottomonium and a stringent test of any description of the heavy quark–antiquark interaction: a model tuned on $c\bar{c}$ and $b\bar{b}$ must interpolate to $c\bar{b}$ with no additional freedom. Experimentally the field has moved quickly. The ground state is known to 0.3 MeV [1]; CMS [2] and LHCb [3] have observed the $B_c(2S)$ and the $B_c^*(2S)$ doublet; lattice QCD predicted the $B_c^*(1S)$ mass to better than 10 MeV [4]; and in 2026 ATLAS observed the $B_c^*(1S) \to B_c\gamma$ transition directly, measuring the hyperfine splitting $m(B_c^*) - m(B_c) = 64.5 \pm 1.4\,^{+1.0}_{-1.4}$ MeV [5]. Excited $P$- and $D$-wave states are within reach of LHCb Run 3.

Non-relativistic potential models [6–8] have long been used to predict the $B_c$ spectrum. Recent work has extended them to higher excitations and radiative decays [9, 10], and has begun to quantify parameter uncertainties with Bayesian methods [9]. A common feature of these studies, however, is that the model parameters are calibrated, at least in part, on the $B_c$ states themselves. This makes agreement with $B_c(1S)$ and $B_c(2S)$ partly circular and obscures the question we wish to address here: *given only the $c\bar{c}$ and $b\bar{b}$ spectra, how well is the $B_c$ spectrum determined, and which of its features are independent of the unknown form of the confining potential?*

We answer this by refitting three potential families to the current $c\bar{c}$ and $b\bar{b}$ data and propagating the fits to $c\bar{b}$ without further adjustment (Secs. 2–4). We then confront the hyperfine sector with three independent measurements: the ATLAS $1S$ splitting, and the $B_c(2S)$–$B_c^*(2S)$ doublet of CMS and LHCb, which determines the *difference* of the $1S$ and $2S$ splittings—a quantity in which the leading systematic uncertainties of the contact interaction cancel (Sec. 5). Radiative widths are given in Sec. 6 and we conclude in Sec. 7.

## 2 Framework

### 2.1 Spin-independent spectrum

For a quark of mass $m_1$ and antiquark of mass $m_2$ interacting through a central potential $V(r)$, the reduced radial wave function $u_{nl}(r) = rR_{nl}(r)$ obeys

$$u'' = 2\mu\Big[V(r) + \frac{l(l+1)}{2\mu r^2} - E_{nl}\Big]u, \qquad \mu = \frac{m_1 m_2}{m_1+m_2}, \tag{1}$$

in natural units. We integrate Eq. (1) with the Numerov algorithm on a uniform grid ($h = 10^{-3}\,\mathrm{GeV}^{-1}$) and locate eigenvalues by bisection on the number of radial nodes; the results are converged to better than $10^{-6}$ GeV with respect to $h$ and the integration range. The spin-averaged mass is $M_{nl} = m_1 + m_2 + E_{nl}$.

We consider three families of potential:

$$V_{\mathrm{C}}(r) = -\frac{\kappa}{r} + \frac{r}{a^2}, \tag{2}$$

$$V_{\mathrm{L}}(r) = D + C\ln r, \tag{3}$$

$$V_\alpha(r) = V_0 + B\,(r^\alpha - 1), \tag{4}$$

with $r$ in $\mathrm{GeV}^{-1}$. Eq. (2) is the Cornell potential [11]; Eq. (3) the Quigg–Rosner logarithmic potential [12]; Eq. (4) is the Martin power law [13] written so that the potential is pinned at $r = 1\,\mathrm{GeV}^{-1}$ ($V_0 = -1.166$ GeV) and the exponent $\alpha$ can be varied without shifting the potential by amounts that the quark masses would otherwise absorb. For $\alpha = 0.1$ it coincides with the Martin potential of Ref. [6]; as $\alpha \to 0$ it tends to Eq. (3) with $C = B\alpha$; $\alpha = 1$ is linear confinement. The additive constants $D$ and $V_0$ are exactly degenerate with $m_1 + m_2$ and are held fixed.

### 2.2 Spin-dependent splittings

Fine and hyperfine structure is added in first-order perturbation theory following Eichten and Quigg [6],

$$\Delta = \sum_{k=1}^{4} T_k, \tag{5}$$

$$T_1 = \Big[\frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle}{2m_1^2} + \frac{\langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{2m_2^2}\Big]\tilde{T}_1, \quad T_2 = \frac{\langle \mathbf{L}\cdot\mathbf{s}_1\rangle + \langle \mathbf{L}\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_2,$$
$$T_3 = \frac{\langle \mathbf{s}_1\cdot\mathbf{s}_2\rangle}{m_1 m_2}\tilde{T}_3, \qquad T_4 = \frac{\langle S_{12}\rangle}{m_1 m_2}\tilde{T}_4, \tag{6}$$

with radial coefficients

$$\tilde{T}_1 = -\Big\langle \frac{1}{r}\frac{dV}{dr}\Big\rangle + 2\tilde{T}_2, \qquad \tilde{T}_2 = \frac{4\alpha_s}{3}\langle r^{-3}\rangle,$$
$$\tilde{T}_3 = \frac{8\alpha_s}{9}|R_{n0}(0)|^2, \qquad \tilde{T}_4 = \frac{\alpha_s}{3}\langle r^{-3}\rangle. \tag{7}$$

The full fitted potential is treated as a Lorentz scalar for the spin–orbit term, while the vector one-gluon exchange enters through $\alpha_s$; this is the prescription of Ref. [6] and keeps the spin-dependent sector free of additional parameters. For $m_1 \neq m_2$ the states with $J = L$ mix; we construct the $2\times 2$ matrix in the $\{^3L_L, {}^1L_L\}$ basis,

$$H = \begin{pmatrix} -\tfrac{1}{2}(a_1+a_2) + 2d & \tfrac{1}{2}(a_1-a_2)\sqrt{L(L+1)} \\ \tfrac{1}{2}(a_1-a_2)\sqrt{L(L+1)} & 0 \end{pmatrix}, \tag{8}$$

with $a_i = \tilde{T}_1/2m_i^2 + \tilde{T}_2/m_1 m_2$ and $d = \tilde{T}_4/m_1 m_2$, and diagonalise it. Its trace and determinant agree with the $jj$-coupling matrices of Ref. [6], Eqs. (2.21). We label the two $J = L$ eigenstates $L^+$ and $L^{+\prime}$ in order of mass.

Two treatments of $\alpha_s$ are compared. In the *fixed* variant $\alpha_s$ is a single fitted constant. In the *running* variant we use the one-loop expression $\alpha_s(\mu) = 12\pi/[25\ln(\mu^2/\Lambda^2)]$ ($n_f = 4$) evaluated at a scale set by the quark masses, either $\mu = 2\mu_{\rm red} = 2m_1 m_2/(m_1 + m_2)$ or $\mu = \sqrt{m_1 m_2}$, and fit $\Lambda$ instead of $\alpha_s$. The two scale choices coincide for $c\bar{c}$ and $b\bar{b}$ and differ only in how the coupling is interpolated to $c\bar{b}$.

### 2.3 Radiative transitions

$E1$ and $M1$ widths are computed from the same wave functions [6],

$$\Gamma_{E1} = \frac{4\alpha\langle e_Q\rangle^2}{27}\, k^3\, (2J_f+1)\, \mathcal{S}_{if}\, |\langle f|r|i\rangle|^2, \tag{9}$$
$$\Gamma_{M1} = \frac{16\alpha}{3}\, \mu_Q^2\, k^3\, (2J_f+1)\, |\langle f|j_0(kr/2)|i\rangle|^2, \tag{10}$$

with $\langle e_Q\rangle = (m_b e_c - m_c e_{\bar{b}})/(m_b+m_c)$, $\mu_Q = (m_b e_c - m_c e_{\bar{b}})/4m_c m_b$, $k = (M_i^2 - M_f^2)/2M_i$, and $\mathcal{S}_{if} = 9\max(L_i, L_f)\{{}^{L_f\ J_f\ S}_{J_i\ L_i\ 1}\}^2$, which reduces to $\mathcal{S} = 1$ for ${}^3S_1 \leftrightarrow {}^3P_J$ and $\mathcal{S} = 3$ for ${}^1P_1 \leftrightarrow {}^1S_0$. For spin-mixed states the $E1$ rate is the sum over the $S = 1$ and $S = 0$ channels weighted by the squared mixing amplitudes. On charmonium this reproduces $\Gamma(\chi_{c0,1,2} \to J/\psi\gamma) = 163, 265, 340$ keV against the measured 151, 288, 374 keV [1].

All code is public (see Code and data availability) and every table and figure in this paper is regenerated by a single script.

Table 1: Experimental input: 27 $J$-resolved levels below open-flavour threshold [1], in MeV. Our radial label $n$ counts excitations from 1, so $\chi_c(1P)$ is $1\,{}^3P_J$. $\Upsilon(4S)$ is excluded.

| System | State | Mass |
|---|---|---|
| $c\bar{c}$ | $1\,{}^1S_0$ | $2984.1 \pm 0.4$ |
| $c\bar{c}$ | $1\,{}^3S_1$ | $3096.9 \pm 0.0$ |
| $c\bar{c}$ | $1\,{}^3P_0$ | $3414.7 \pm 0.3$ |
| $c\bar{c}$ | $1\,{}^3P_1$ | $3510.7 \pm 0.1$ |
| $c\bar{c}$ | $1\,{}^1P_1$ | $3525.4 \pm 0.1$ |
| $c\bar{c}$ | $1\,{}^3P_2$ | $3556.2 \pm 0.1$ |
| $c\bar{c}$ | $2\,{}^1S_0$ | $3637.7 \pm 0.9$ |
| $c\bar{c}$ | $2\,{}^3S_1$ | $3686.1 \pm 0.0$ |
| $c\bar{c}$ | $1\,{}^3D_1$ | $3773.7 \pm 0.7$ |
| $c\bar{c}$ | $1\,{}^3D_2$ | $3823.5 \pm 0.5$ |
| $c\bar{c}$ | $1\,{}^3D_3$ | $3842.7 \pm 0.2$ |
| $b\bar{b}$ | $1\,{}^1S_0$ | $9398.7 \pm 2.0$ |
| $b\bar{b}$ | $1\,{}^3S_1$ | $9460.4 \pm 0.1$ |
| $b\bar{b}$ | $1\,{}^3P_0$ | $9859.4 \pm 0.5$ |
| $b\bar{b}$ | $1\,{}^3P_1$ | $9892.8 \pm 0.4$ |
| $b\bar{b}$ | $1\,{}^1P_1$ | $9899.3 \pm 0.8$ |
| $b\bar{b}$ | $1\,{}^3P_2$ | $9912.2 \pm 0.4$ |
| $b\bar{b}$ | $2\,{}^1S_0$ | $9999.0 \pm 3.5$ |
| $b\bar{b}$ | $2\,{}^3S_1$ | $10023.4 \pm 0.5$ |
| $b\bar{b}$ | $1\,{}^3D_2$ | $10163.7 \pm 1.4$ |
| $b\bar{b}$ | $2\,{}^3P_0$ | $10232.5 \pm 0.6$ |
| $b\bar{b}$ | $2\,{}^3P_1$ | $10255.5 \pm 0.6$ |
| $b\bar{b}$ | $2\,{}^1P_1$ | $10259.8 \pm 1.2$ |
| $b\bar{b}$ | $2\,{}^3P_2$ | $10268.6 \pm 0.6$ |
| $b\bar{b}$ | $3\,{}^3S_1$ | $10355.1 \pm 0.5$ |
| $b\bar{b}$ | $3\,{}^3P_1$ | $10513.4 \pm 0.7$ |
| $b\bar{b}$ | $3\,{}^3P_2$ | $10524.0 \pm 0.8$ |

## 3 Fit to charmonium and bottomonium

Table 1 lists the input. We fit each family's potential parameters, $m_c$, $m_b$ and $\alpha_s$ (or $\Lambda$) by least squares to all 27 levels, using $\sigma^2 = \sigma_{\rm exp}^2 + \sigma_{\rm th}^2$ with a theory floor $\sigma_{\rm th} = 10$ MeV so that the keV-precision $J/\psi$ and $\psi(2S)$ masses do not dominate a model whose intrinsic accuracy is tens of MeV. Parameter uncertainties are taken from the covariance of the fit and later propagated linearly to predictions.

The results are in Table 2 and Fig. 1. Residuals are typically 10–30 MeV and are largest for the charmonium $D$ waves (all families miss $\psi_3(3842)$ by $\sim 40$ MeV), which lie at the $D\bar{D}$ threshold where coupled-channel effects are expected. Compared with the 1994 parameters [6], the logarithmic and Martin families change little except for $\alpha_s$, which rises from 0.37–0.43 to 0.45–0.46; the Cornell quark masses, by contrast, drop by 0.4–0.5 GeV, confirming that the

Table 2: Fitted parameters and $\chi^2$ per degree of freedom for a 10 MeV theory floor. Units: $B$, $C$, $m_c$, $m_b$, $\Lambda$ in GeV; $a$ in GeV$^{-1}$. For the Martin family $B$, $m_c$ and $m_b$ are strongly correlated (see text); the individual errors are large but predicted masses are not.

| Family | $\chi^2$/dof | Parameters |
|---|---|---|
| Martin | 85.0/22 | $B = 7.19 \pm 0.45$, $\alpha = 0.096(6)$, $m_c = 1.63 \pm 0.25$, $m_b = 5.02 \pm 0.23$, $\alpha_s = 0.458(26)$ |
| Log | 73.8/23 | $C = 0.737(5)$, $m_c = 1.509(4)$, $m_b = 4.908(2)$, $\alpha_s = 0.449(22)$ |
| Cornell | 134.1/22 | $\kappa = 0.455(10)$, $a = 2.328(20)$, $m_c = 1.334(9)$, $m_b = 4.752(7)$, $\alpha_s = 0.374(19)$ |
| Martin, running $\alpha_s$ | 69.8/22 | $B = 6.84 \pm 0.50$, $\alpha = 0.101(7)$, $m_c = 1.83 \pm 0.27$, $m_b = 5.20 \pm 0.26$, $\Lambda = 0.431(94)$ |
| Log, running $\alpha_s$ | 59.0/23 | $C = 0.740(5)$, $m_c = 1.509(4)$, $m_b = 4.905(2)$, $\Lambda = 0.334(25)$ |

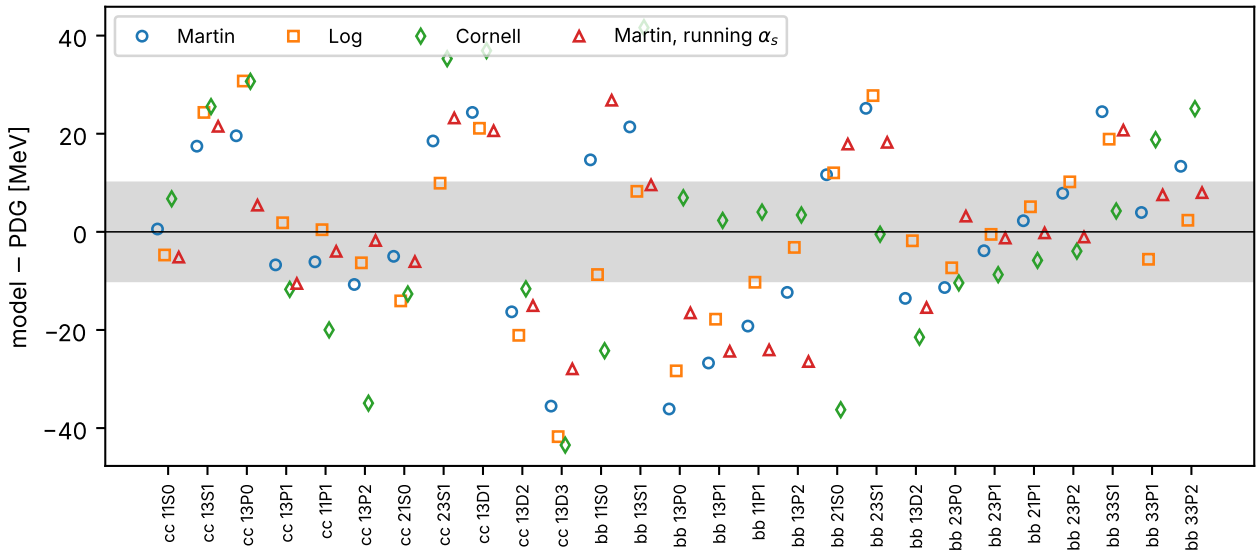



Figure 1: Residuals (model − PDG) of the fits for the three potential families and the running-coupling Martin fit. The grey band is the 10 MeV theory floor.

original Cornell masses were tuned to a different mass convention. The running coupling improves the fit in both families it was applied to ($\chi^2$: 85 → 70 for Martin, 74 → 59 for logarithmic) because it allows the $c\bar{c}$ and $b\bar{b}$ hyperfine splittings, 113 and 62 MeV, to be reproduced simultaneously; the fitted $\Lambda = 0.33$–$0.43$ GeV corresponds to $\alpha_s(m_c) \simeq 0.50$ and $\alpha_s(m_b) \simeq 0.28$–$0.30$.

For the Martin family the parameters $B$, $m_c$ and $m_b$ carry large formal errors because $B(r^\alpha - 1) \simeq B\alpha \ln r$ for small $\alpha$, so $B$ and the masses are nearly degenerate. The combination $m_c + m_b$ that enters $B_c$ is nevertheless well constrained, and the propagated uncertainties on predicted masses are $\lesssim 15$ MeV.

To probe the shape of the confining potential we repeat the fit of Eq. (4) at fixed $\alpha \in \{0.05, 0.1, 0.2, 0.35, 0.5, 0.75, 1\}$ (Table 3, Fig. 2). The $\chi^2$ rises monotonically from 73 at $\alpha = 0.05$ to 986 at $\alpha = 1$: the $c\bar{c}$ and $b\bar{b}$ spectra together strongly prefer a nearly logarithmic potential, consistent with the classic observation that the $2S$–$1S$ spacing is almost flavour independent [12]. A free fit of $\alpha$ returns 0.096(6).

## 4 The $B_c$ spectrum

Table 4 gives the $B_c$ spectrum through $3S$, $2P$ and $1D$ predicted by the three families, and Fig. 3 shows the dependence on $\alpha$ across the scan. Three features stand out.

First, the $S$-wave levels are remarkably stable. $B_c(1S)$, $B_c^*(1S)$ and $B_c(2S)$ vary by $\pm 14$, $\pm 17$ and $\pm 6$ MeV across the three families and by less than 70 MeV over the whole $\alpha$ scan, even though $\chi^2$ changes by an order of magnitude. The ground state is reproduced at $6276 \pm 14$ MeV against the measured 6274.5 MeV without any $B_c$ input.

Second, the excited $P$, $D$ and $3S$ levels are not stable: the $1P$ centroid moves by 80 MeV and the $3\,^3S_1$ by 150 MeV between $\alpha = 0$ and $\alpha = 1$, and the Cornell family—which the $c\bar{c} + b\bar{b}$ fit disfavours—sits 30–60 MeV above the others for the $2P$ and $3S$. These are the states for which future measurements will discriminate between confinement models; restricted to the $\alpha \lesssim 0.2$ region preferred by $c\bar{c} + b\bar{b}$, the $1P$ centroid is 6737–6768 MeV and the $1D$ centroid 7023–7043 MeV.

Third, the spin-averaged $B_c(2S)$ is predicted 11–24 MeV *above* the measured 6871.2 MeV in every one of the thirteen fits performed in this work (Tables 4 and 3), with a propagated uncertainty of only 2–

Table 3: Fits at fixed confining power $\alpha$ and the resulting $B_c$ levels (MeV). Centroids are spin-weighted averages of the fine-structure multiplets. The last column is the predicted $\Delta_{1S}-\Delta_{2S}$; experiment gives $29.1\pm1.7$ MeV [2]. $\alpha=0$ denotes the logarithmic fit.

| $\alpha$ | $\chi^2$/dof | $\alpha_s$ | $B_c(1S)$ | $B_c^*(1S)$ | $B_c(2S)$ | $1P$ | $1D$ | $3\,^3S_1$ | $\Delta_{1S}-\Delta_{2S}$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 74/23 | 0.449 | 6266 | 6349 | 6889 | 6768 | 7043 | 7253 | 40.4 |
| 0.05 | 73/23 | 0.462 | 6257 | 6337 | 6878 | 6752 | 7030 | 7254 | 37.3 |
| 0.1 | 85/22 | 0.458 | 6267 | 6343 | 6886 | 6753 | 7034 | 7270 | 33.8 |
| 0.2 | 136/23 | 0.466 | 6269 | 6338 | 6882 | 6737 | 7023 | 7285 | 27.4 |
| 0.35 | 254/23 | 0.466 | 6280 | 6340 | 6886 | 6724 | 7014 | 7312 | 19.4 |
| 0.5 | 404/23 | 0.464 | 6292 | 6344 | 6889 | 6713 | 7006 | 7336 | 13.1 |
| 0.75 | 689/23 | 0.458 | 6309 | 6351 | 6893 | 6698 | 6990 | 7371 | 5.4 |
| 1 | 986/23 | 0.453 | 6325 | 6360 | 6897 | 6686 | 6976 | 7400 | 0.0 |

Table 4: $B_c$ spectrum (MeV) predicted from the $c\bar{c}+b\bar{b}$ fits with fixed $\alpha_s$. Parenthesised errors are the propagated fit uncertainties; the model spread is half the range over the three families. The comparison columns are Eichten and Quigg (Buchmüller–Tye potential) [6] and PDG 2024 [1]. Our $1P$ and $1D$ correspond to the $2P$ and $3D$ of the principal-quantum-number labelling.

| State | Martin | Log | Cornell | mean $\pm$ spread | EQ94 | PDG |
|---|---|---|---|---|---|---|
| $1\,^1S_0$ | 6267(11) | 6266(4) | 6295(4) | $6276\pm14$ | 6264 | $6274.5\pm0.32$ |
| $1\,^3S_1$ | 6343(10) | 6349(4) | 6376(4) | $6356\pm17$ | 6337 | $6339.0\pm1.7$ |
| $1\,^3P_0$ | 6692(15) | 6711(4) | 6724(4) | $6709\pm16$ | 6700 | – |
| $1^+$ | 6735(13) | 6753(3) | 6753(3) | $6747\pm9$ | 6730 | – |
| $1^{+\prime}$ | 6756(10) | 6770(3) | 6759(2) | $6762\pm7$ | 6736 | – |
| $1\,^3P_2$ | 6774(10) | 6787(3) | 6770(3) | $6777\pm9$ | 6747 | – |
| $2\,^1S_0$ | 6886(11) | 6889(2) | 6877(3) | $6884\pm6$ | 6856 | $6871.2\pm1.0$ |
| $2\,^3S_1$ | 6928(10) | 6931(2) | 6930(3) | $6930\pm1$ | 6899 | – |
| $1\,^3D_1$ | 7029(14) | 7040(3) | 7036(3) | $7035\pm5$ | 7012 | – |
| $1D_2$ | 7029(9) | 7037(3) | 7022(3) | $7029\pm7$ | 7009 | – |
| $1D_2'$ | 7040(14) | 7050(3) | 7044(3) | $7044\pm5$ | 7012 | – |
| $1\,^3D_3$ | 7035(9) | 7043(3) | 7027(3) | $7035\pm8$ | 7005 | – |
| $2\,^3P_0$ | 7101(13) | 7102(4) | 7148(5) | $7117\pm23$ | 7108 | – |
| $1^+$ | 7132(12) | 7131(3) | 7176(4) | $7146\pm23$ | 7135 | – |
| $1^{+\prime}$ | 7151(10) | 7147(3) | 7185(4) | $7161\pm19$ | 7142 | – |
| $2\,^3P_2$ | 7164(10) | 7159(3) | 7196(4) | $7173\pm19$ | 7153 | – |
| $3\,^1S_0$ | 7239(11) | 7225(4) | 7290(6) | $7251\pm32$ | 7244 | – |
| $3\,^3S_1$ | 7270(10) | 7253(4) | 7335(6) | $7286\pm41$ | 7280 | – |

11 MeV. Since the $c\bar{c}$ and $b\bar{b}$ $2S$ levels are themselves reproduced to $\lesssim 25$ MeV with residuals of both signs, this is a genuine interpolation failure at the few-MeV level rather than a fit artefact, and is the most robust discrepancy we find. We return to it in Sec. 7.

Our fixed-$\alpha_s$ $1P$ and $1D$ levels lie 5–30 MeV above those of Ref. [6], whose Buchmüller–Tye potential is softer at the relevant distances; the ordering within each multiplet and the mixing of the $J=L$ states (mixing angle $\simeq 23^\circ$ for the $1P$) are the same.

# 5 The hyperfine test

In both the CMS and LHCb analyses the $B_c^*(2S)$ is reconstructed through $B_c^*(2S)\to B_c^*(1S)\pi\pi$, $B_c^*(1S)\to B_c\gamma$ with the soft photon unobserved, so the observed peak sits at $M(B_c^*(2S))-\Delta_{1S}$, where $\Delta_{nS}\equiv M(n^3S_1)-M(n^1S_0)$. The measured separation of the two peaks is therefore

$$M(B_c(2S))-M_{\rm peak}=\Delta_{1S}-\Delta_{2S}, \qquad (11)$$

$=29.1\pm1.5\pm0.7$ MeV (CMS [2]) and 30.9 MeV (LHCb [3]). This is a quantity in which potential models are unusually trustworthy. From Eq. (7),

$$\Delta_{1S}-\Delta_{2S}=\frac{8\alpha_s}{9m_cm_b}\big[|R_{10}(0)|^2-|R_{20}(0)|^2\big], \qquad (12)$$

so the overall normalisation uncertainties of the contact interaction—the value of $\alpha_s$, relativistic corrections to $|R(0)|^2$, the choice of quark masses—enter as a common factor, while the shape of the potential enters through the ratio $|R_{20}(0)|^2/|R_{10}(0)|^2$.

Table 5 and Fig. 4 show that the three fixed-$\alpha_s$ families predict $\Delta_{1S}-\Delta_{2S}=28$–40 MeV, bracketing the measurement, and that across the $\alpha$ scan (Fig. 2, Table 3) the prediction falls smoothly from 37 MeV at $\alpha=0.05$ to zero at $\alpha=1$, where $|R_{n0}(0)|^2$ is independent of $n$. The measurement therefore selects $\alpha\simeq0.1$–

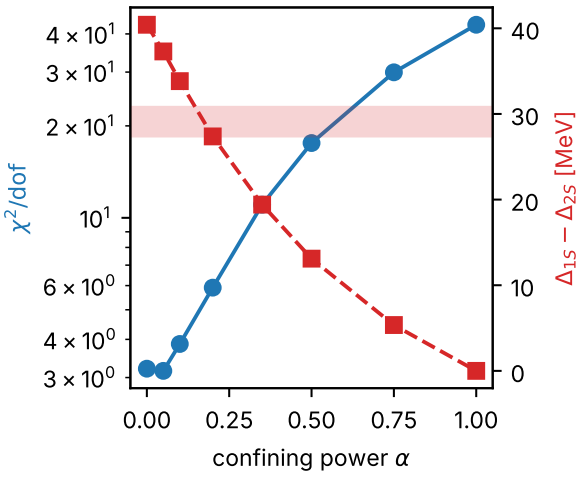



Figure 2: Left axis: $\chi^2$/dof of the $c\bar{c} + b\bar{b}$ fit as a function of the confining power $\alpha$ in Eq. (4). Right axis: the predicted hyperfine difference $\Delta_{1S} - \Delta_{2S}$ of $B_c$; the band is the CMS measurement [2].

Table 5: Hyperfine splittings (MeV) of $B_c$ predicted from $c\bar{c} + b\bar{b}$, for fixed and running $\alpha_s$. Experiment: $\Delta_{1S} = 64.5 \pm 1.7$ and $M(B_c^*) = 6339.0 \pm 1.7$ (ATLAS [5]); $\Delta_{1S} - \Delta_{2S} = 29.1 \pm 1.7$ (CMS [2]) and 30.9 (LHCb [3], statistical error $\approx 1.5$); hence $\Delta_{2S} = 35.4 \pm 2.4$. Lattice QCD (2010): $M(B_c^*) = 6330 \pm 9$ [4].

| Family | $\Delta_{1S}$ | $\Delta_{2S}$ | $\Delta_{1S} - \Delta_{2S}$ | $M(B_c^*)$ |
|---|---|---|---|---|
| Martin | 75.9 | 42.1 | 33.8 | 6343 |
| Log | 82.5 | 42.1 | 40.4 | 6349 |
| Cornell | 81.2 | 53.1 | 28.2 | 6376 |
| Martin, running ($\sqrt{m_c m_b}$) | 62.9 | 35.1 | 27.9 | 6331 |
| Martin, running ($2\mu$) | 67.4 | 37.6 | 29.9 | 6332 |
| Log, running ($\sqrt{m_c m_b}$) | 66.5 | 33.9 | 32.6 | 6342 |
| Log, running ($2\mu$) | 72.1 | 36.8 | 35.3 | 6344 |

Table 6: $E1$ widths above 10 keV and $M1$ widths of the lowest $B_c$ states, for the Martin fit with fixed $\alpha_s$. $k$ is the photon energy. Spin-mixed states are treated as described in Sec. 2.

| Transition | Type | $k$ [MeV] | $\Gamma$ [keV] |
|---|---|---|---|
| $1^{+\prime} \to 1\,^1S_0\,\gamma$ | E1 | 472 | 135.2 |
| $1\,^3P_2 \to 1\,^3S_1\,\gamma$ | E1 | 417 | 110.7 |
| $1\,^3D_1 \to 1\,^3P_0\,\gamma$ | E1 | 329 | 94.6 |
| $1\,^3D_3 \to 1\,^3P_2\,\gamma$ | E1 | 256 | 80.5 |
| $1^+ \to 1\,^3S_1\,\gamma$ | E1 | 381 | 71.2 |
| $1^{+\prime} \to 2\,^1S_0\,\gamma$ | E1 | 260 | 65.6 |
| $1\,^3P_0 \to 1\,^3S_1\,\gamma$ | E1 | 340 | 59.9 |
| $2\,^3P_2 \to 2\,^3S_1\,\gamma$ | E1 | 232 | 57.1 |
| $1D_2' \to 1^+\,\gamma$ | E1 | 298 | 52.5 |
| $1D_2 \to 1^{+\prime}\,\gamma$ | E1 | 267 | 46.6 |
| $1D_2' \to 1^{+\prime}\,\gamma$ | E1 | 278 | 46.2 |
| $1D_2 \to 1^+\,\gamma$ | E1 | 287 | 42.4 |
| $1\,^3D_1 \to 1^+\,\gamma$ | E1 | 288 | 40.2 |
| $1^+ \to 2\,^3S_1\,\gamma$ | E1 | 201 | 30.4 |
| $2\,^3P_0 \to 2\,^3S_1\,\gamma$ | E1 | 172 | 23.0 |
| $1^+ \to 1\,^1S_0\,\gamma$ | E1 | 452 | 21.6 |
| $1^{+\prime} \to 1\,^1S_0\,\gamma$ | E1 | 830 | 18.7 |
| $2\,^3P_2 \to 1\,^3S_1\,\gamma$ | E1 | 774 | 18.6 |
| $1^{+\prime} \to 1\,^3S_1\,\gamma$ | E1 | 401 | 15.1 |
| $2\,^3S_1 \to 1\,^3P_2\,\gamma$ | E1 | 152 | 14.9 |
| $2\,^3P_0 \to 1\,^3S_1\,\gamma$ | E1 | 718 | 14.9 |
| $2\,^3S_1 \to 1^+\,\gamma$ | E1 | 190 | 14.7 |
| $2\,^1S_0 \to 1^{+\prime}\,\gamma$ | E1 | 128 | 13.6 |
| $1^+ \to 1\,^3S_1\,\gamma$ | E1 | 746 | 13.6 |
| $2\,^3P_2 \to 1\,^3D_3\,\gamma$ | E1 | 128 | 12.8 |
| $1^+ \to 2\,^1S_0\,\gamma$ | E1 | 242 | 11.9 |
| $1D_2' \to 1\,^3P_2\,\gamma$ | E1 | 261 | 11.5 |
| $2\,^3S_1 \to 1\,^3P_0\,\gamma$ | E1 | 232 | 10.6 |
| $1\,^3S_1 \to 1\,^1S_0\,\gamma$ | M1 | 75 | 0.1 |
| $2\,^3S_1 \to 1\,^1S_0\,\gamma$ | M1 | 629 | 0.1 |
| $2\,^1S_0 \to 1\,^3S_1\,\gamma$ | M1 | 521 | 0.1 |
| $2\,^3S_1 \to 2\,^1S_0\,\gamma$ | M1 | 42 | 0.0 |

0.2, in agreement with—and independent of—the preference of the $c\bar{c} + b\bar{b}$ spectrum itself. Two unrelated observables point to the same confining potential.

The absolute splitting is a sharper test still, now that ATLAS has measured it directly: $\Delta_{1S} = 64.5 \pm 1.7$ MeV [5], some 9 MeV (one standard deviation) above the 2010 lattice prediction [4]. Combined with the CMS difference this fixes $\Delta_{2S} = 35.4 \pm 2.4$ MeV, so the two lowest hyperfine splittings of $B_c$ are both known to a few MeV. Figure 5 and Table 5 compare them with the predictions. With fixed $\alpha_s$ all three families give $\Delta_{1S} = 76$–$82$ MeV and $\Delta_{2S} = 42$–$53$ MeV, 10–20 MeV too high, even though each reproduces the $c\bar{c}$ and $b\bar{b}$ $1S$ splittings to $\lesssim 20$ MeV: a single coupling cannot interpolate the contact term $\propto \alpha_s |R(0)|^2/m_1 m_2$ between flavours. With the one-loop running coupling the Martin family gives $\Delta_{1S} = 63$–$67$ MeV, $\Delta_{2S} = 35$–$38$ MeV and $M(B_c^*) = 6331$–$6332$ MeV, and the logarithmic family $\Delta_{1S} = 67$–$72$ MeV, $\Delta_{2S} = 34$–$37$ MeV—both within or close to the ATLAS and CMS uncertainties. We stress that $\Lambda$ in these fits is determined by the $c\bar{c}$ and $b\bar{b}$ hyperfine splittings alone, and that neither a fixed coupling nor any adjustment of the spin-independent potential can reproduce the measured values. A one-parameter running coupling fitted to equal-mass quarkonia therefore predicts the unequal-mass hyperfine sector of $B_c$—the absolute $1S$ splitting, the $2S$ splitting and their difference—without $B_c$ input. The same fits place the lattice value [4] at the low edge of the allowed range; an updated lattice determination of $m(B_c^*)$ with current precision would be valuable.

## 6 Radiative transitions

Table 6 lists the $E1$ and $M1$ widths of the $1P$, $1D$ and $2P$ states. The $1P \to 1S$ widths of 60–135 keV agree in magnitude with Refs. [6, 10]; the spin mixing of the $1^+$ states redistributes the rate so that the heavier $1^{+\prime}$ decays mainly to $B_c\gamma$ (135 keV) and the lighter $1^+$ mainly to $B_c^*\gamma$ (71 keV). The $1D \to 1P$ widths of 40–95 keV make the cascade $1D \to 1P\gamma \to 1S\gamma\gamma$

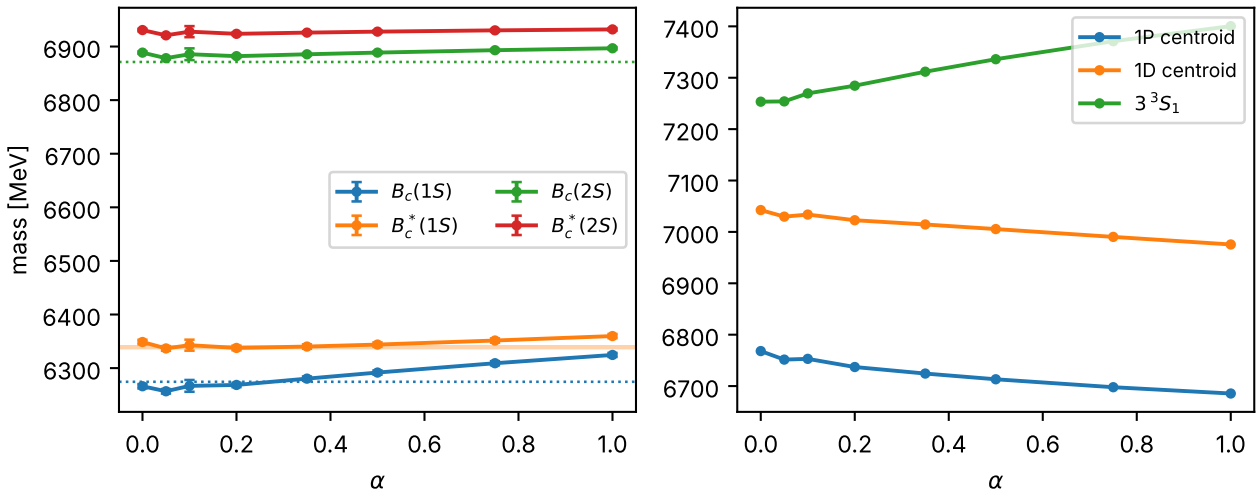



Figure 3: $B_c$ levels as a function of the confining power $\alpha$ of the potential fitted to $c\bar{c} + b\bar{b}$. Left: $S$-wave states; dotted lines are the measured $B_c(1S)$ and $B_c(2S)$, the band is the ATLAS $B_c^*(1S)$ [5]. Right: $1P$ and $1D$ centroids and the $3\,^3S_1$.

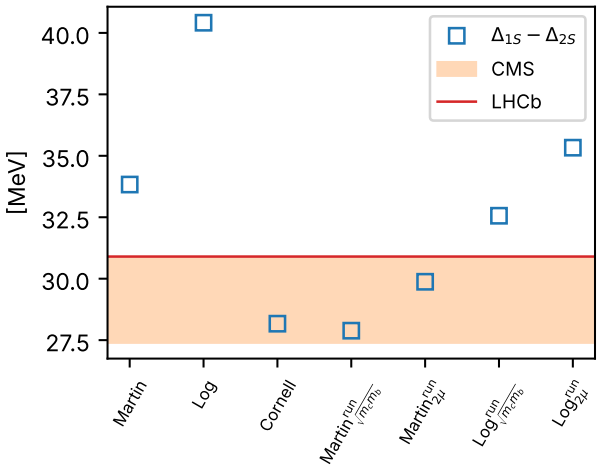



Figure 4: Predicted $\Delta_{1S} - \Delta_{2S}$ for each fit, compared with CMS (band) and LHCb (line).

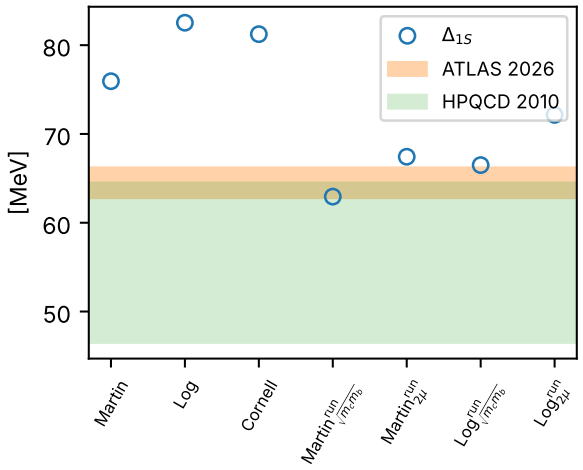



Figure 5: Predicted absolute $1S$ hyperfine splitting $\Delta_{1S}$ for each fit, compared with the ATLAS measurement [5] (narrow band) and the 2010 lattice prediction [4] (wide band).

the natural route to the first $D$-wave $B_c$ state. The $M1$ rates are $\lesssim 0.1$ keV; the $B_c^*(1S) \to B_c\gamma$ photon ($k \simeq 60$–$75$ MeV) remains the principal obstacle to a direct $B_c^*$ mass measurement. We have not included the model dependence of these widths, which is dominated by the photon energy ($\propto k^3$) and is therefore largest for transitions involving the excited levels of Sec. 4.

## 7 Conclusions

Fitting three families of potentials to the 2024 charmonium and bottomonium spectra and propagating them to $c\bar{b}$ without further input, we find:

1. The $c\bar{c} + b\bar{b}$ data prefer a nearly logarithmic confining potential, $\alpha \lesssim 0.1$, and the $S$-wave $B_c$ levels are determined to $\pm 15$ MeV independently of the potential; the $P$, $D$ and $3S$ levels are not, and are the states to measure.

2. The hyperfine difference $\Delta_{1S} - \Delta_{2S}$ measured by CMS and LHCb is predicted correctly from $c\bar{c}+b\bar{b}$ and independently selects $\alpha \simeq 0.1$–$0.2$.

3. A fixed $\alpha_s$ overpredicts the $B_c^*(1S)$ hyperfine splitting measured by ATLAS by 10–20 MeV; a one-loop running coupling, with $\Lambda$ fixed by $c\bar{c}$ and $b\bar{b}$, predicts $\Delta_{1S} = 63$–$72$ MeV and $\Delta_{2S} = 34$–$38$ MeV, in agreement with ATLAS, CMS and LHCb simultaneously.

4. The spin-averaged $B_c(2S)$ is predicted 11–24 MeV too high in every variant.

The last point deserves comment. The $2S$ discrepancy survives changes of confining potential, of $\alpha_s$ treatment and of theory floor, and is several times the propagated uncertainty. It is the first level at which the first-order, non-relativistic treatment used here appears to fail for $B_c$ while succeeding for $c\bar{c}$ and $b\bar{b}$, and we suggest that it reflects either relativistic corrections that do not interpolate linearly in $1/m$, or the proximity of the $2S$ to the $BD$ threshold. A relativised treatment [7] or a coupled-channel calculation with the $c\bar{c} + b\bar{b}$-determined potential of this work would settle the question.

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## Code and data availability

The complete code, the fitted parameter sets and the scripts that regenerate every table and figure in this paper are available at `https://github.com/apoppong24/bc-quarkonium`; the experimental inputs are listed in Table 1.

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